

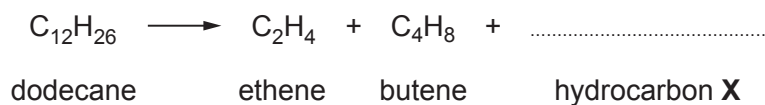
6. (a) Kerosene is one of the fractions separated during the fractional distillation of crude oil.

Kerosene contains dodecane, $C_{12}H_{26}$. Dodecane undergoes a further process **A** to form smaller, more useful hydrocarbons, ethene, butene and one molecule of hydrocarbon **X**.

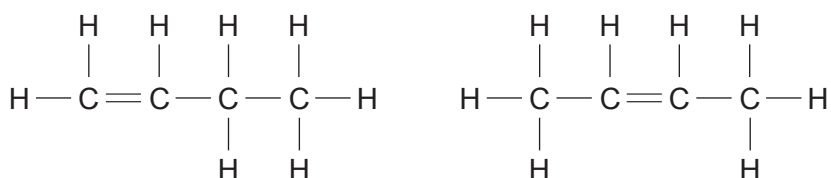
- (i) Name process **A**. [1]

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- (ii) Complete the equation for this reaction. [1]



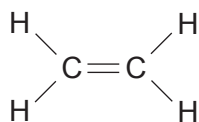
- (b) C_4H_8 has three isomers. The diagrams below show two of the three isomers.



- The third isomer is methylpropene. Draw its structure. [1]



- (c) (i) Polyethene is made from ethene by addition polymerisation. Draw the structure of the repeating unit for polyethene. [1]

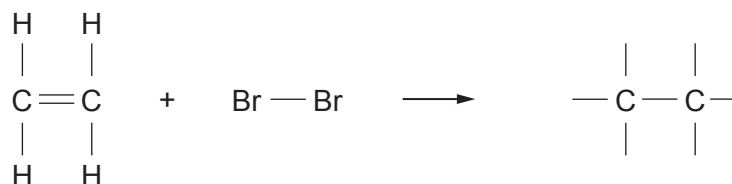


ethene

repeating unit for polyethene

- (ii) The presence of the double bond in ethene can be confirmed using bromine water.

- I. Complete the equation for the reaction between ethene and bromine. [1]



- II. Tick (✓) the box next to the name of the product formed. [1]

1,2-dibromoethene

☐

1,1-dibromoethane

☐

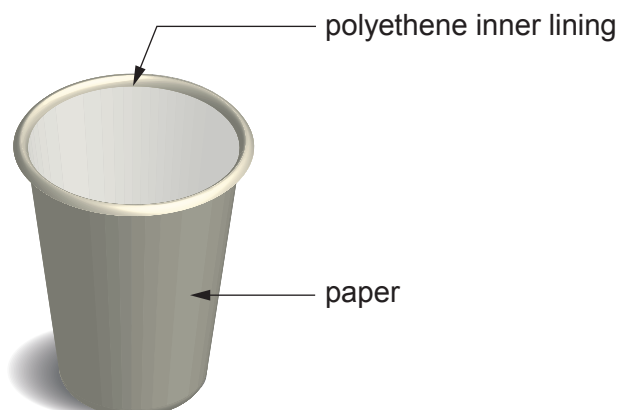
1,2-dibromoethane

☐

1,1-dibromoethene

☐


- (d) In the UK, around 3 billion disposable coffee cups are used every year. Each paper coffee cup is lined with polyethene which is impossible to remove at a recycling plant. If the automated machine that sorts waste at a recycling centre detects a plastic lining, it rejects the cup sending it to general household waste.



- (i) Most general household waste is disposed of in landfill sites.

Apart from using landfill sites, give the **other** disposal method of general household waste. State an environmental problem associated with the method. [2]

Method

Problem

- (ii) Recycling helps to conserve raw materials. Name the raw material used to make polyethene. Give the main reason why it is important to conserve this raw material. [2]

Raw material

Main reason

